

# The Data End of Crypto Quant for Public LLMs: An Observation Layer that Compiles World Bundles

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## Abstract

**Purpose.** End-to-end crypto quant for public LLMs needs a *data end* before decision or execution. We build that layer as an anonymized observation runtime: multi-band evidence  $\rightarrow$  BandPIT/WMI  $\rightarrow$  a point-in-time (PIT) *world bundle*, with hard refusal when support is thin. **Evidence.** Primary—seeing: grounding recovers ready/missing/tilt at  $\approx 0.97$  under Compiled vs.  $\lesssim 0.23$  under Raw, while sufficiency chat alone is unreliable (Compiled mean 0.60). Secondary—valve: under a *frozen full-schema* stress gate (max WMI  $\approx 0.093 < 0.2$ ), Compiled thin-abstain is 1.0 (100-day and full OOS  $n=2000$ ) while Ungated soft judgment is only  $\approx 0.68$  /  $\approx 0.56$ ; Raw over-trades. Secondary—handoff: the *scoped-archive production valve* opens on  $\approx 61\%$  OOS days; a no-LLM tilt rule then acts (Sharpe 0.77, CE +0.25). **Claim.** Build the quant data end first: compile so models can see; refuse when they must not act.

## CCS Concepts

• **Computing methodologies**  $\rightarrow$  **Artificial intelligence**; *Machine learning*; • **Applied computing**  $\rightarrow$  *Economics*.

## Keywords

data end, observation layer, world bundle, grounding, public LLMs, cryptocurrency, point-in-time, quality-gated refusal, systems

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## 1 Introduction

*Purpose (read this first).* Build the *data end* of end-to-end crypto quant for public LLMs. The stack is **data/observation**  $\rightarrow$  **decision**  $\rightarrow$  **execution** (Table 1); agents and generative world models [6, 7, 9, 19–21] typically assume the first stage. Crypto violates that assumption: exchange, macro, and alternative evidence arrive on incompatible clocks [11, 13]. We own only the data end—compile a PIT world bundle so models can *see*, and hard-refuse when the view is too thin. Refusal is a safety valve, not the product and not a trading claim.

*Failure modes.* No feed  $\rightarrow$  refuse (safe, useless as a data end). Thin price fragment  $\rightarrow$  trade and lose (unsafe for any downstream stack). Disclosed but ungated world  $\rightarrow$  read fields, disagree on action. So the data end must (i) make the market legible and (ii) enforce a quality valve before decision/execution.

**Table 1: End-to-end quant stack; this paper owns only the data end.**

| Stage              | Typical owner            | This paper                   |
|--------------------|--------------------------|------------------------------|
| Data / observation | Assumed by agents        | World bundle + quality valve |
| Decision           | LLM / policy / portfolio | Out of scope                 |
| Execution          | Broker / OMS             | Out of scope                 |

*What we build.* An anonymized runtime (Section 4): multi-band evidence  $\rightarrow$  BandPIT/WMI  $\rightarrow$  JSON world bundle for GPT / DeepSeek / GLM / Gemini class models, served through a read API a decision layer can call. “World model” means state compiler + quality gate (Section 3), not dynamics and not a decision engine. The bundle is complete, honest, and auditable; `should_ai_abstain` is the machine-readable valve.

*How we evaluate (dual protocol).* **RQ1 (primary—seeing):** Are ready / missing / tilt answers verifiable against PIT ground truth under Compiled vs. Raw? **RQ2 (valve):** Under a *frozen full-schema stress gate* (WMI over all schema bands; never clears 0.2 on this panel), does a hard flag enforce thin abstention where soft disclosure does not? Four arms: Compiled, Ungated (same content, no flag), Raw (mom5), Blind. **RQ3a (handoff):** Under the *scoped-archive production valve* (declared bands only; open iff archive-complete and WMI  $\geq 0.2$ ), can a transparent no-LLM rule act when the valve opens? **RQ3b (content):** On the same vintaged tilts, does an ungated content probe beat momentum ( $\Delta$ CE; LOBO)?

*Contributions.* (1) Data-end semantics: epistemic observations, compilation  $\Pi_t$ , world-quality index with band scope, and seeing vs. gating with a lag-missingness bound. (2) Anonymized PIT compile path: collectors  $\rightarrow$  BandPIT  $\rightarrow$  service surface (bundle / health / hand-off reads)  $\rightarrow$  world bundles; frozen four-arm harness. (3) Live + hand-off validation: grounding  $\approx 0.97$ ; full-schema stress thin-abstain 1.0; scoped Compiled-open handoff (open  $\approx 0.61$ , Sharpe 0.77); ungated tilt contrast ( $\Delta$ CE=0.334; LOBO content share 1.0). Estimand: data-end legitimacy, valve reliability, and handoff usability.

## 2 Related Work

*World models.* World models learn compact states and often dynamics [6, 7, 9]. We address the complementary observation-side problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy—not a Dreamer/JEPA simulator.

*LLM agents in finance.* Return prediction [12], open financial LLMs [18], memory- and tool-augmented agents [10, 19–21], and ICAIF retrieval/debate/judge systems [1, 3, 15, 16] improve how agents fetch, debate, and are scored. They still assume a usable observation stream at the data end of the stack. We study that prior stage: compile a world bundle, measure whether models can *see* it, then gate action when they must not act. Our estimand is therefore orthogonal to memory routers and debate judges: without a typed

PIT world, those stacks consume whatever fragment the tool dump provides.

*Selective prediction and PIT.* Abstention can be optimal under noise [4, 5]; we attach refusal to world quality  $q(W_t)$  via a runtime field, not classifier confidence alone (Prop. 3.8). Macro vintages [2] and crypto microstructure [11, 13] motivate multi-band compilation with disclosed missingness (Prop. 3.6). Bootstrap discipline for the content probe follows [8, 14, 17].

### 3 Observation-Layer Semantics

This section states the data-end theory as *interface semantics*: what counts as an observation, how compilation yields a world, and when quality forces refusal. It is not a pricing theorem and not a generative dynamics model. Systems realization is Section 4; empirical tests are Section 5.

*Definition 3.1 (Epistemic observation).*  $O_{j,t} = (x_{j,t}, \tau_{j,t}, q_{j,t}, g_{j,t}, r_{j,t})$ : value, latest-available time, quality, main-view gate, semantic role. Omitting  $(\tau, q, g, r)$  yields a feature dump, not a world observation.

*Definition 3.2 (Compilation).*  $\Pi_t = B_t \circ M_t \circ A_t$ : align clocks, apply honesty/missingness gates, assemble band views into a consumer object.

*Definition 3.3 (World bundle).*  $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$  with scalar quality  $q(W_t)$ ;  $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$ .  $W_t$  is the handoff object to decision.

*Definition 3.4 (World-quality index).* Ready/limited/missing counts disclose coverage.  $\text{WMI}_t = B_t U_t H_t$  aggregates breadth, stability, and honesty in  $[0, 1]$ ; ACWMI multiplies a regime weight. Band scope selects the support of  $(B, U, H)$ : *full schema* (stress / ceiling audit) vs. the *declared consumer archive* (production valve). Scoped WMI scores only declared bands; it is a contract-completeness index, not a lowered  $q^*$ . The Compiled interface exposes `should_ai_abstain` (and, under archive scope, incompleteness) with `thin_world` (production  $q^* = 0.2$  for scoped WMI).

*Seeing vs. gating.* The bundle is what the model *sees*; the hard flag *blocks* action when  $q(W_t) < q^*$ . Tilts without a gate are a dump; a gate without a readable bundle is only a switch. The two interfaces are separable: Ungated keeps  $W_t$  and numeric  $q(W_t)$  but removes the hard boolean.

*Assumption 3.5 (Bounded lag and increments).* Band  $j$  has max observation lag  $\Delta_j$ ; latent increments satisfy  $\|S_u - S_{u-1}\| \leq \delta$ .

**PROPOSITION 3.6 (LAG-MISSINGNESS RECONSTRUCTION BOUND).** *Under Assumption 1, any compiler that uses only in-clock observations satisfies*

$$\|\tilde{S}_t - S_t\| \leq C_{\text{del}} \delta \sum_j \Delta_j + C_{\text{noi}} \epsilon_t + C_{\text{miss}} m_t,$$

where  $m_t$  is disclosed missing mass and  $\epsilon_t$  collects measurement noise. Operationally, BandPIT maps age to ready/limited/missing so  $m_t$  and lag terms are readable in  $W_t$ ; compilation cannot erase delay, but it can refuse to pretend the bound is zero.

**PROPOSITION 3.7 (COMPILATION  $\neq$  DUMP).** *Enlarging raw span without  $\Pi_t$  need not enlarge usable  $\mathcal{F}^{\text{AI}}$ : ungated stale evidence can raise volume while lowering honesty, tightening rather than loosening Prop. 3.6.*

**PROPOSITION 3.8 (WORLD-CONDITIONAL ABSTENTION).** *Let  $c_{\text{act}}(W_t)$  be the expected cost of non-abstain actions and  $c_{\text{abs}}$  the abstain cost. If  $c_{\text{act}}(W_t) > c_{\text{abs}}$  and  $c_{\text{act}}$  is nonincreasing in  $q(W_t)$ , the Bayes action is abstain on a lower set  $\{W : q(W) < q^*\}$  [4]. A hard runtime flag implements that set; soft text does not.*

**PROPOSITION 3.9 (LOBO CONTENT VS. GATING).** *Deleting band  $j$  changes a downstream value functional through a content channel and a gating channel. Under an ungated rule the gating channel is shut, so the value gap identifies nonempty fields.*

*Theory  $\rightarrow$  protocol.* Prop. 3.6–3.7  $\Rightarrow$  RQ1 (recover disclosed readiness/tilts). Prop. 3.8  $\Rightarrow$  RQ2 (hard flag vs. soft judgment under the full-schema stress gate). Def. 3.4 (band scope) + Prop. 3.8  $\Rightarrow$  RQ3a (scoped production valve opens; handoff acts). Prop. 3.9  $\Rightarrow$  RQ3b (ungated content / LOBO share).

**REMARK 3.10 (NOT A GENERATIVE WM).** *We do not learn  $p(s_{t+1} | s_t, a_t)$ . We deliver a state compiler with a quality-gated refuse interface—the data end of the stack.*

## 4 Runtime: Data-End Compile Path

The anonymized prototype implements the data end as a layered compile path (Figure 1, Table 2). Decision and execution sit downstream; this paper does not claim them. Paths and product names are omitted for double-blind review.

### 4.1 Layered design

- (1) **Data layer.** Exchange, macro, and alternative collectors write vintage-aware history (observation timestamps; macro `available_at`).
- (2) **Store layer.** Embedded stores hold raw series, merged market panels, and compiled analytics (readiness, WMI/ACWMI, world snapshots).
- (3) **Logic layer.** BandPIT reconstructs readiness on a previous-close clock; honesty/missingness gates implement  $\Pi_t$ ; WMI/ACWMI and availability shocks  $O_t$  (band outage events for quasi-exogenous levers) are first-class.
- (4) **Service layer.** A read API serves quality-tagged bundles and health objects without restarting collectors for newly arrived rows. Anonymized surface includes world-bundle reads, `/bundle/{symbol}/handoff` (valve-gated tilt action, no LLM), health exposing `should_ai_abstain` with `band_scope`, a paper world-model time slice, and availability-shock queries.
- (5) **Consumer layer.** Frozen-prompt LLM/rule agents call an OpenAI-compatible adapter at temperature 0 under four arms.

### 4.2 Evaluation archive

The runtime schema is band-extensible; this study evaluates the three-band consumer object in Table 3 on a PIT panel of  $\approx 3990$  asset-days (2025-07 to 2026-08): exchange ready share  $\approx 0.80$  panel-wide ( $\approx 0.61$  OOS), macro 1.0, alternative 1.0. **Full-schema stress WMI** scores permanently empty non-archive slots and caps at max  $\approx 0.093$ , so a 0.2 gate never opens—the frozen LLM-arm protocol (RQ2). **Scoped-archive production WMI** renormalizes  $B, U, H$  to {exchange, macro, alternative} and opens iff the archive is complete

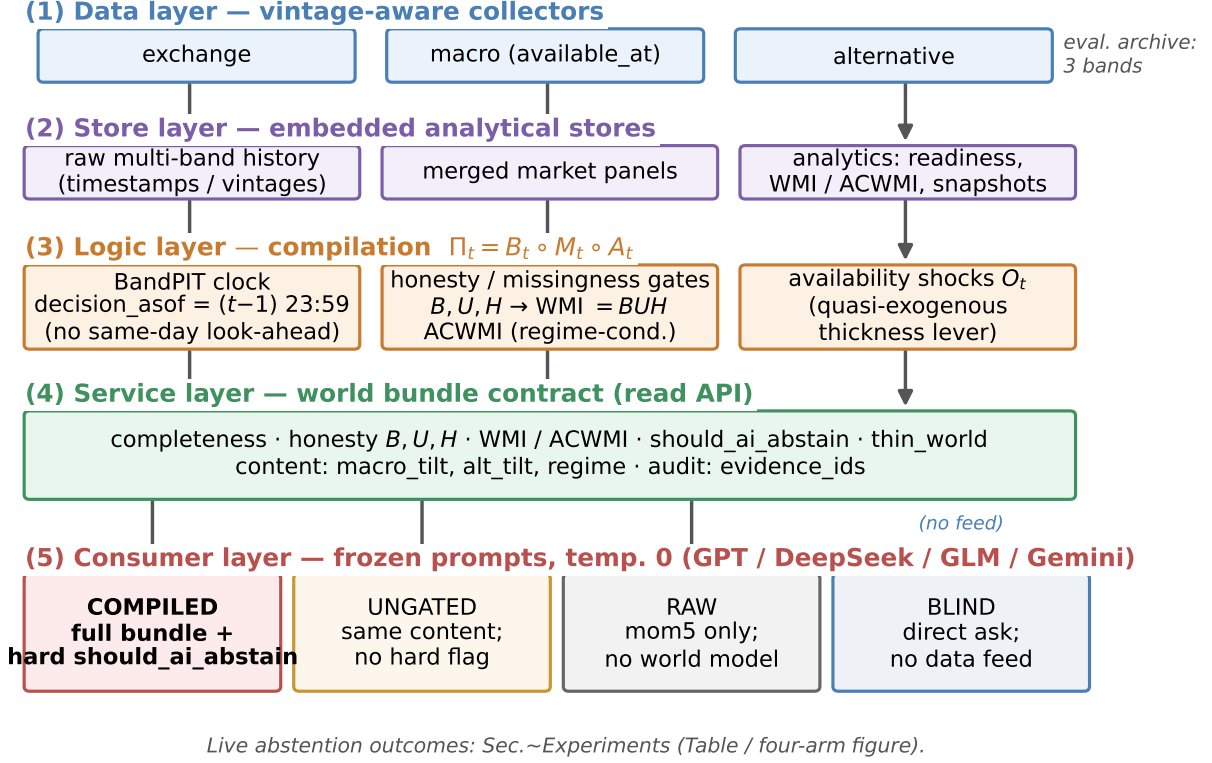


Figure 1: Data-end pipeline: evidence → BandPIT → world bundle → Compiled / Ungated / Raw / Blind.

Table 2: Compile-path modules (systems view, not a product inventory).

| Module              | Role                                                   |
|---------------------|--------------------------------------------------------|
| Collectors / stores | Timestamped / vintaged band history                    |
| BandPIT compiler    | Previous-close readiness; $\Pi_t$                      |
| Quality indices     | $B, U, H \rightarrow WMI$ ; band scope; abstain; $O_t$ |
| Bundle builder      | Complete / honest / auditable JSON                     |
| Service surface     | Bundle / health / handoff / availability reads         |
| Consumer harness    | Four-arm protocol; temp. 0                             |

Table 3: Input evidence vs. fields emitted in the world bundle (live archive).

| Band        | Fetches (raw)               | Emitted (bundle)                             |
|-------------|-----------------------------|----------------------------------------------|
| exchange    | OHLCV, funding, OI          | mom5; status/age                             |
| macro       | equity-vol / dollar indices | macro_tilt; status/age                       |
| alternative | stablecoin circulation      | alt_tilt; status/age                         |
| derived     | —                           | WMI, ACWMI, abstain, $O_t$ ,<br>evidence IDs |

and  $WMI \geq 0.2$  (RQ3a); this measures contract readiness, not a relaxed threshold.

### 4.3 PIT clock and freshness

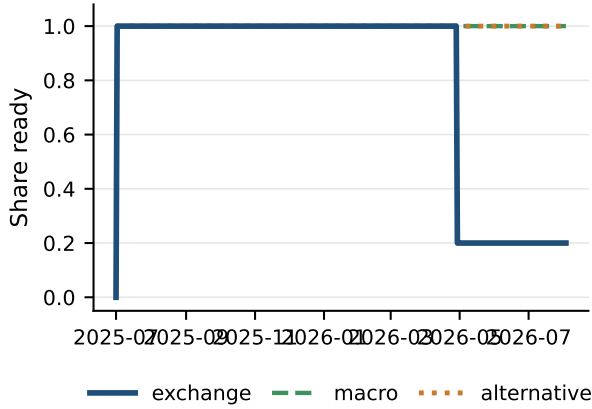
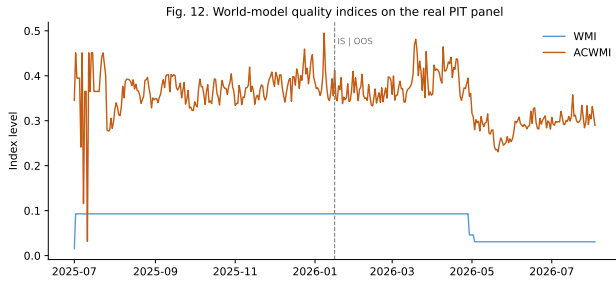
For calendar day  $t$ , BandPIT reconstructs the decision information set at  $(t-1)$  23:59 (payoff is same-day close-to-close return); the bundle records decision\_asof. Each band is graded ready /limited /missing against daily-scale age thresholds (Table 4). Missingness is disclosed rather than imputed (Prop. 3.6). Breadth, stability, and honesty aggregate into  $WMI_t = B_t U_t H_t$  (ACWMI adds regime weight). PIT-safe tilts use only in-clock observations: five-day macro changes, a seven-day stablecoin-flow slope, and mom5. The *scoped production valve* sets should\_ai\_abstain when the archive is incomplete or scoped  $WMI < 0.2$ , and exposes thin\_world; bundles also disclose full\_schema\_wmi as a ceiling audit. Figure 2: macro and alternative stay ready; exchange readiness varies and defines thick/thin days. Figure 3: full-schema WMI stays under 0.2; ACWMI can look healthier under regime weight; the runtime exports both indices and band\_scope. Table 5: scoped production valve opens on  $\approx 61\%$  OOS days ( $\equiv 3/3$  ready); full-schema stress gate stays closed.

### 4.4 World bundle (what the LLM sees)

The consumer object is a JSON world bundle (Table 6): timing, completeness, honesty, quality (including the abstain flag), content tilts, and evidence\_ids. Listing 1: exchange-gap day (scoped valve closed). Listing 2: 3/3-ready day where scoped  $WMI=1.0$  opens the

**Table 4: BandPIT freshness thresholds (seconds → days).**

| Band                      | Fresh window | In eval archive             |
|---------------------------|--------------|-----------------------------|
| exchange                  | 2d           | yes (ready $\approx 0.80$ ) |
| macro                     | 5d           | yes (ready 1.0)             |
| alternative               | 7d           | yes (ready 1.0)             |
| news / on-chain / options | 3d           | missing                     |
| tokenomics                | 7d           | missing                     |
| event calendar            | 14d          | missing                     |

**Figure 2: PIT readiness: what is visible in the archive over time.****Figure 3: Full-schema stress paths: WMI stays under 0.2; ACWMI is regime-weighted (ceiling audit).****Table 5: Dual gates vs. open share (OOS). Production = scoped; stress = full schema.**

| Gate                   | Open share | Note                             |
|------------------------|------------|----------------------------------|
| Scoped production 0.2  | 0.61       | complete $\wedge$ WMI $\geq 0.2$ |
| Full-schema stress 0.2 | 0.00       | max WMI $\approx 0.093$          |

production valve; full\_schema\_wmi  $\approx 0.093$  is disclosed as ceiling audit—same day, two scopes, not a lowered  $q^*$ .

**Listing 1: Exchange-gap bundle (Compiled; scoped closed).**

```
{ "date": "2026-05-11", "asset": "ADA",
```

**Table 6: World-bundle field groups (the market view).**

| Group        | Fields                                      |
|--------------|---------------------------------------------|
| Timing       | decision_asof, previous-close protocol      |
| Completeness | ready/limited/missing counts, band status   |
| Honesty      | $B, U, H$ ; main-view / stale gates         |
| Quality      | WMI, ACWMI, band_scope, abstain, thin_world |
| Content      | macro_tilt, alt_tilt, mom5                  |
| Audit        | evidence_ids; EAR; full_schema_wmi          |

```
"decision_asof": "2026-05-10T23:59:00",
"completeness": {"n_ready": 2, "n_missing": 1},
"world_model_index": {"band_scope": "eval_archive",
  "wmi": 0.139, "archive_complete": false,
  "should_ai_abstain": true, "thin_world": true,
  "full_schema_wmi": 0.015},
"macro_tilt": 1.0, "alt_tilt": -1.0}
```

**Listing 2: 3/3-ready day: scoped valve opens.**

```
{ "date": "2026-01-16", "asset": "ADA",
  "completeness": {"n_ready": 3, "n_missing": 0},
  "world_model_index": {"band_scope": "eval_archive",
    "wmi": 1.0, "archive_complete": true,
    "should_ai_abstain": false, "thin_world": false,
    "full_schema_wmi": 0.093},
  "band_status": {"exchange": "ready", "macro": "ready",
    "alternative": "ready"},
  "macro_tilt": 1.0, "alt_tilt": -1.0}
```

## 4.5 Four arms and frozen prompts

Actions  $\in$  {bullish, bearish, neutral, abstain}. **Compiled**: full bundle; frozen prompt *requires* abstain when should\_ai\_abstain is true (and when the world is judged thin). **Ungated**: same content, completeness, and numeric WMI; no hard flag; soft prompt asks the model to judge thinness. **Raw**: mom5 only—thin price wiring for the ladder. **Blind**: “Today is  $d$ . How should I trade  $a$ ?” with no feed. **Dual protocol**. Live LLM arms (RQ1–RQ2) freeze *full-schema* bundles (max WMI  $\approx 0.093$ ), so Compiled abstains every OOS day—a closed stress test of the hard flag. RQ3a uses the *scoped production valve* (default band scope) on the same panel: open on band-thick days, then hand off without an LLM. Compiled vs. Ungated differ only in enforcement (Listings 3–4); prompts, IS/OOS cut, and seeds are frozen. On thin Compiled days the evidence-alignment proxy (EAR) is 1.0 across vendors: rationales stay bound to disclosed evidence\_ids.

**Listing 3: Compiled (excerpt): hard valve.**

1. If should\_ai\_abstain is true OR the world is thin, you MUST “abstain”.
2. Prefer band content over raw momentum (macro\_tilt, alt\_tilt are PIT-safe).
3. Rationale must match evidence\_ids.

**Listing 4: Ungated (excerpt): soft judgment.**

1. Judge whether low WMI / missingness makes action unsafe; you MAY “abstain”.
  2. Prefer band content when you act.
- NO hard abstain boolean; NO forced policy.

*Why the hard valve is not a confound.* Compiled abstention is partly prompt obedience, and that is intentional: the data end’s job on thin days is to *block* action, not to hope the model discovers thinness (Prop. 3.8). Ungated measures soft judgment from the same

disclosed bundle; Raw measures fragment wiring; Blind measures no feed. The gating result is that the hard flag is reliable across vendors; soft refusal is not—which is why half-baked context remains dangerous without the valve.

#### 4.6 Systems validation (engines)

Paper-lab smoke wires BandPIT/WMI/ACWMI and checks the dual gate: scoped archive opens on 3/3-ready statuses while full-schema stays closed; a handoff rule acts only when the scoped valve is open. Availability shocks  $O_t$  are queryable on the service surface.

### 5 Experiments

#### 5.1 Setup

PIT panel  $\sim 399$  days  $\times 10$  liquid names ( $\sim 3990$  asset-days; 2025-07–2026-08), chronological IS/OOS 200/200 days (cut 2026-01-16). Unless noted, live arms share the same stratified 100 OOS asset-days (seed fixed), temperature 0, and four flash/mini vendors (gpt-5.4-mini, deepseek-v4-flash, glm-5.2, gemini-3.5-flash-lite) via an OpenAI-compatible adapter. Grounding uses 50 OOS days; full-OOS abstention uses  $n=2000$ ; no-LLM probes use  $\approx 200$  OOS days after return alignment. Primary metrics: RQ1 grounding F1/tilt accuracy (sufficiency chat secondary); RQ2 thin-world abstain under the *full-schema stress gate* (CE as avoided loss, not alpha); RQ3a scoped Compiled-open handoff Sharpe/CE; RQ3b ungated tilt  $\Delta$ CE vs. momentum.

#### 5.2 RQ1 (primary): Seeing—grounding workflow

If the data end works, models must report a *checkable* market view—not invent coverage from a price fragment.

On 50 OOS asset-days, each model lists ready/missing bands among exchange/macro/alternative, reports macro\_tilt / alt\_tilt signs, and judges sufficiency; (i)–(iii) score against PIT ground truth.

Table 7: Compiled mean ready/missing F1 and tilt-sign accuracy 0.97; Raw collapses to 0.04, 0.23, 0.19. Raw has no readiness or tilt fields, so it cannot recover archive state; Compiled forces use of the disclosed world.

*Legibility  $\neq$  sufficiency chat.* Table 8: Compiled sufficiency accuracy averages only 0.60 (Gemini 0.04) despite 0.97 field grounding, while Raw often answers “enough?” correctly by chance/heuristics (0.95) while inventing coverage. Chat judgment of “is this enough to trade?” is not a substitute for a typed world; we therefore measure seeing by checkable fields (RQ1) and action safety by the hard valve (RQ2).

*Case study.* Listings 5–6: ADA (2026-05-11) exchange missing; Compiled recovers the checkable view; Raw cannot. A second day (2026-01-17) under the *full-schema stress* Compiled arm abstains on should\_ai\_abstain; Raw goes bullish from mom5 = +1 alone.

#### Listing 5: Compiled grounding (gpt-5.4-mini, ADA 2026-05-11).

```
ready: alternative, macro
missing: exchange
macro_tilt +; alt_tilt -; sufficiency: insufficient
"thin world... should_ai_abstain=true"
```

**Table 7: Grounding (50 days): primary seeing result. C = Compiled; R = Raw.**

| Model                 | Ready F1 |      | Missing F1 |      | Tilt-sign |      |
|-----------------------|----------|------|------------|------|-----------|------|
|                       | C        | R    | C          | R    | C         | R    |
| gpt-5.4-mini          | 0.94     | 0.00 | 0.94       | 0.31 | 0.94      | 0.20 |
| deepseek-v4-flash     | 0.98     | 0.06 | 0.98       | 0.20 | 0.98      | 0.18 |
| glm-5.2               | 0.98     | 0.09 | 1.00       | 0.11 | 0.98      | 0.20 |
| gemini-3.5-flash-lite | 0.96     | 0.03 | 0.96       | 0.30 | 0.96      | 0.18 |
| Mean                  | 0.97     | 0.04 | 0.97       | 0.23 | 0.97      | 0.19 |

**Table 8: Sufficiency chat vs. field grounding (50 days). Seeing = fields; chat alone misleads.**

| Model                 | Suff. C | Suff. R | Ready F1 C | Ready F1 R |
|-----------------------|---------|---------|------------|------------|
| gpt-5.4-mini          | 0.78    | 0.98    | 0.94       | 0.00       |
| deepseek-v4-flash     | 0.78    | 0.90    | 0.98       | 0.06       |
| glm-5.2               | 0.80    | 0.96    | 0.98       | 0.09       |
| gemini-3.5-flash-lite | 0.04    | 0.94    | 0.96       | 0.03       |
| Mean                  | 0.60    | 0.95    | 0.97       | 0.04       |

**Table 9: Thin abstention on identical 100 OOS days under full-schema stress (Wilson 95% CIs). Blind = 1.0.**

| Model                 | Thin C | Thin U [CI]  | Abs. R [CI]  | $\Delta$ CE |
|-----------------------|--------|--------------|--------------|-------------|
| gpt-5.4-mini          | 1.00   | 0.63 [53,72] | 0.04 [02,10] | +0.73       |
| deepseek-v4-flash     | 1.00   | 0.81 [72,87] | 0.75 [66,82] | +1.22       |
| glm-5.2               | 1.00   | 0.86 [78,91] | 0.11 [06,19] | +1.04       |
| gemini-3.5-flash-lite | 1.00   | 0.43 [34,53] | 0.07 [03,14] | +1.30       |
| Mean                  | 1.00   | 0.68         | —            | +1.07       |

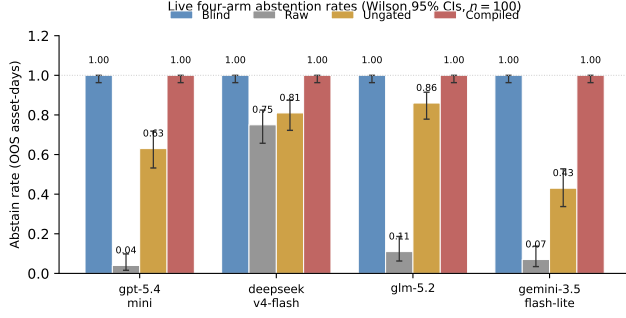
#### Listing 6: Raw grounding / action (same model family).

```
grounding: ready=[]; cannot confirm tilts
action (2026-01-17): bullish
"Positive 5-period momentum in raw feed."
```

#### 5.3 RQ2 (safety valve): Hard refuse when the view is thin

Legibility does not imply safe action. All live LLM arms in this subsection freeze *full-schema* bundles, so every OOS day is thin under the stress gate 0.2. Table 9 and Figure 4: on the same 100 days, Compiled abstains on every thin day (1.0); Ungated soft-judges at mean 0.68 (Gemini 0.43); Raw over-trades (Table 10); Blind abstains 1.0 with no feed—safe but useless as a data end. The costly failure mode is the middle case: half-baked Raw context makes models act and lose. Mean  $\Delta$ CE (Compiled–Raw) +1.07 is avoided loss under the stress valve, not an alpha claim.

*Full-OOS replication.* Table 11: on all 2000 OOS asset-days (all thin under the *full-schema stress gate* 0.2), Compiled thin-abstain remains 1.000 (Wilson [.998, 1]) for every vendor. Ungated soft rates fall to a four-vendor mean  $\approx 0.56$ ; Raw GPT/Gemini abstain at 0.000. Scale does not rescue soft judgment. Blind refuses everywhere with no feed on the paired 100-day sample (abstain 1.0 for every vendor):



**Figure 4: Safety valve: Blind refuses; Raw over-trades; Ungated is unstable; Compiled enforces the hard gate.**

**Table 10: Raw action mix (100 days): half-baked context drives directional mass.**

| Model                 | Abs. | Bull. | Bear. | Neut. |
|-----------------------|------|-------|-------|-------|
| gpt-5.4-mini          | 0.04 | 0.28  | 0.38  | 0.30  |
| deepseek-v4-flash     | 0.75 | 0.14  | 0.04  | 0.07  |
| glm-5.2               | 0.11 | 0.40  | 0.48  | 0.01  |
| gemini-3.5-flash-lite | 0.07 | 0.44  | 0.49  | 0.00  |

**Table 11: Full-OOS thin abstention ( $n=2000$ ; all thin under full-schema stress gate 0.2).**

| Model                 | Thin C | Thin U         | Abs. R |
|-----------------------|--------|----------------|--------|
| gpt-5.4-mini          | 1.000  | 0.666          | 0.000  |
| deepseek-v4-flash     | 1.000  | 0.540          | 0.748  |
| glm-5.2               | 1.000  | 0.679*         | 0.078* |
| gemini-3.5-flash-lite | 1.000  | 0.359          | 0.000  |
| Mean                  | 1.000  | $\approx 0.56$ | —      |

\*GLM coverage incomplete (Ungated  $n=1402$ ; Raw  $n=675$ ).

the failure mode that needs a data end is not “models never abstain,” but “thin fragments make them act.”

*Inside Ungated soft acts.* On the mixed 100-day sample, when Ungated *does* act, act-conditional CE is sharply negative for three of four vendors (DeepSeek  $-3.12$ , GLM  $-4.28$ , Gemini  $-1.67$ ) while GPT is positive on its smaller act set. Soft disclosure without a hard flag is not a free lunch: the same readable world that RQ1 scores can still harm decisions when the valve is removed.

*Open slice (still under full-schema LLM arms).* Table 12: on the band-thick subset (3/3 ready  $\equiv$  full-schema WMI  $\geq 0.05$ ;  $\approx 61\%$  OOS;  $n \approx 1224$ ) Ungated keeps compiled content without the hard boolean—mean abs.  $\approx 0.48$ , CE positive for every vendor (mean  $\approx +0.29$ ; bootstrap  $\approx +0.31$ , CIs include 0). Full-schema Compiled still abstains at 1.0 here; Raw stays risky (GPT/Gemini abs. 0.0, CE negative). The same thick days are where the *scoped production valve* opens; Compiled-open handoff is RQ3a (no new LLM quota).

**Table 12: Open slice under full-schema LLM arms (3/3 ready;  $n \approx 1224$ ). Soft-judgment lower bound.**

| Model                 | Abs. U | CE U  | Act-CE U | Abs. R | CE R   |
|-----------------------|--------|-------|----------|--------|--------|
| gpt-5.4-mini          | 0.58   | +0.42 | +0.32    | 0.00   | −0.28  |
| deepseek-v4-flash     | 0.42   | +0.21 | +0.45    | 0.75   | +0.04  |
| glm-5.2               | 0.58   | +0.24 | +0.11    | 0.09*  | −0.87* |
| gemini-3.5-flash-lite | 0.34   | +0.31 | +0.06    | 0.00   | −0.25  |

\*Raw GLM  $n=398$ ; Ungated GLM  $n=932$ ; others  $n=1224$ . Bootstrap mean CE  $\approx +0.31$  ( $B=5,999$  reps; one-sided  $p \approx 0.21$ ).

**Table 13: RQ3a handoff (no LLM). Scoped production valve vs. full-schema stress.**

| Policy (reads bundle fields)  | Sharpe | CE <sup>†</sup> | Abstain |
|-------------------------------|--------|-----------------|---------|
| Full-schema stress 0.2        | 0.000  | 0.000           | 1.00    |
| Scoped production 0.2 + tilts | 0.772  | +0.253          | 0.39    |
| Momentum (mom5)               | 0.101  | −0.075          | 0.00    |
| Buy-and-hold                  | −1.399 | −0.987          | 0.00    |

<sup>†</sup>CE = ann. mean  $- \frac{1}{2}$  ann. variance (valve-aware path).

**Table 14: RQ3b ungated content probe (no valve). Bootstrap  $\Delta$ CE 0.334.**

| Policy             | Sharpe | CE <sup>‡</sup>     |
|--------------------|--------|---------------------|
| Buy-and-hold       | −1.404 | −1.163              |
| Momentum           | 0.096  | −0.206              |
| World-bundle tilts | 0.764  | +0.130              |
| Tilts – Momentum   | —      | 0.334 ( $p=0.034$ ) |

<sup>‡</sup>Content-harness CE (always-on; not comparable 1:1 to <sup>†</sup>).

## 5.4 RQ3a (handoff): Scoped Compiled-open proof

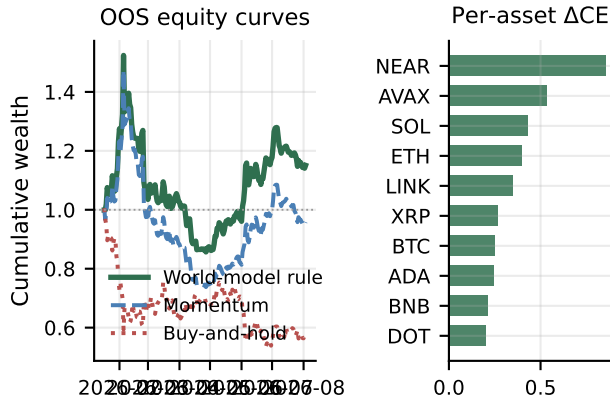
The data end must be usable by a downstream decision rule without an LLM in the loop. The full-schema *stress* gate never clears 0.2, so it cannot demonstrate handoff—by design in RQ2. The scoped *production* valve opens on  $\approx 61\%$  OOS days ( $\equiv$  3/3 ready); a bundle-reading handoff rule abstains when the valve is closed and otherwise acts on disclosed macro\_tilt / alt\_tilt (API: world-bundle handoff read). Table 13: scoped Compiled-open Sharpe 0.772 / CE +0.253 (abstain 0.39) vs. momentum and buy-and-hold; full-schema stress remains fully closed.

## 5.5 RQ3b (content): Nonempty fields

Table 14 and Figure 5: ungated always-on tilts vs. momentum on the same vintaged fields ( $\Delta$ CE=0.334,  $p=0.034$ ; LOBO content share 1.0). This probe shuts the gating channel (Prop. 3.9); it is not the production valve.

*Density split and LOBO channel.* Table 15: open/thick  $\Delta$ CE=0.315; on exchange-gap days the always-on probe still trades (high slice CE is a selection artifact, not a valve claim;  $\Delta$ CE=0.129). Table 16 and Figure 6 (Prop. 3.9): deleting macro or alternative collapses the gap through the content channel (share 1.0; gating residual 0).





**Figure 5: Secondary content probe: compiled tilts beat momentum;  $\Delta$ CE positive for every asset.**

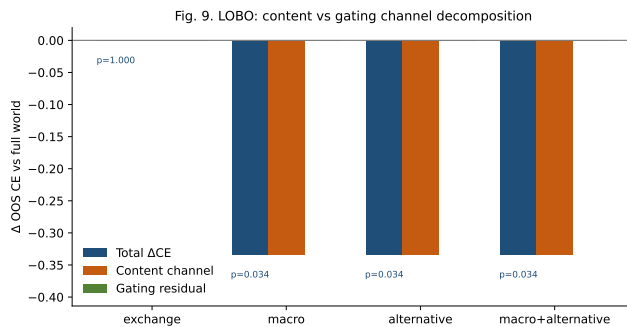
**Table 15: RQ3b content probe by archive density (OOS; ungated).**

| Slice               | Share | Mech CE            | $\Delta$ CE vs mom |
|---------------------|-------|--------------------|--------------------|
| All OOS             | 1.00  | 0.130              | 0.334*             |
| Open / thick        | 0.61  | 0.061              | 0.315              |
| Thin (exchange gap) | 0.39  | 1.048 <sup>§</sup> | 0.129              |

\*Block-bootstrap  $\Delta$ CE; raw mean gap 0.336. <sup>§</sup>Always-on selection artifact.

**Table 16: LOBO decomposition (ungated content probe). Content share 1.0.**

| Drop        | $\Delta$ CE content | $p$   | Content share |
|-------------|---------------------|-------|---------------|
| macro       | -0.334              | 0.034 | 1.0           |
| alternative | -0.334              | 0.034 | 1.0           |
| exchange    | 0.000               | 1.000 | —             |



**Figure 6: LOBO: deleting macro/alternative removes the edge via content, not the refuse flag.**

## 6 Discussion

*What carries the paper.* (i) The data end is legible: grounding  $\approx 0.97$ , with sufficiency chat a poor substitute (RQ1 / Props. 3.6–3.7).

(ii) Under the full-schema stress gate, soft text fails at scale; Blind refuses with no feed; Raw acts on fragments and loses; the hard flag works (RQ2 / Prop. 3.8). (iii) Scoped production WMI opens on band-thick days; a no-LLM handoff acts on disclosed tilts (RQ3a / Def. 3.4+Prop. 3.8); ungated tilts are nonempty (RQ3b / Prop. 3.9; LOBO share 1.0). Relative to FinMem /FinAgent and ICAIF agent stacks [1, 3, 15, 16, 20, 21], we supply the observation stage those systems assume.

*Architecture.* Semantics (Section 3) define the interfaces; Runtime (Section 4) implements  $\Pi_t$  and the service handoff; Experiments (Section 5) test seeing, stress gating, then scoped handoff and content in that order. Trading PnL is intentionally secondary: it confounds observation quality with policy skill.

*Design.* World bundle not feature dump; previous-close PIT with explicit freshness; four arms separate seeing+gating, seeing without gating, thin wiring, and no feed. Dual WMI scope separates stress audit from production readiness; health/bundle/handoff reads make both callable.

*Reproducibility.* Paper-lab smoke checks BandPIT/WMI/ACWMI and scoped-vs-full gates; PIT rebuild and the frozen four-arm harness regenerate the panel and live transcripts used here. Prompts, IS/OOS cut, seeds, and vendor IDs are fixed; credentials stay in environment variables. An anonymized harness ships on acceptance.

*Scope.* We evaluate the data end on the PIT archive above; decision and execution sit downstream.

## 7 Conclusion

End-to-end crypto quant for public LLMs needs a reliable data end before decision or execution. We contribute that observation layer: compile asynchronous evidence into a consumable PIT world bundle, disclose structural missingness, and hard-refuse when the view is too thin. Grounding recovers ready/missing/tilt at  $\approx 0.97$  under Compiled vs. Raw collapse; sufficiency chat alone fails; the full-schema stress gate abstains 1.0 where soft judgment does not; the scoped production valve opens Compiled handoff on band-thick days (Sharpe 0.77). Build the quant data end first: compile so models can see; refuse when they must not act.

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